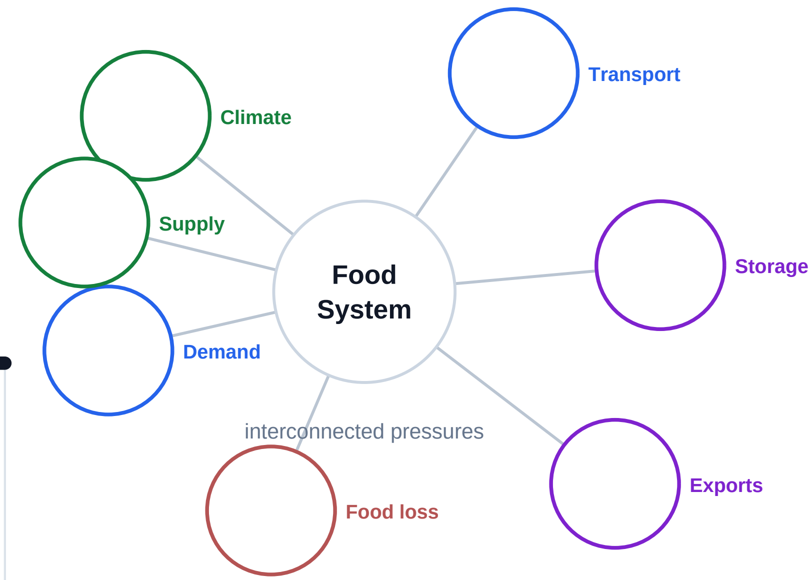
ALGEBRA 2 CUMULATIVE SEMESTER FINAL PROJECT

Systems Under Pressure

The Mathematics of Food Systems



"All models are wrong, but some are useful."

George Box

A spacious mathematical field notebook for sketching, modeling, revising, and explaining.

JOURNAL NORMS

Rough work belongs here.

This journal is not a worksheet packet. It is the visible record of your group's evolving mathematical thinking.

Value uncertainty

A strong investigation often begins confused. Label assumptions, questions, disagreements, and places where a model stops making sense.

Revise visibly

Cross out, annotate, draw arrows, compare versions, and explain why your thinking changed.

Models are arguments

A model is not an answer. It is a claim built from assumptions, evidence, representations, and limits.

Field note style

Messy is acceptable. Hidden thinking is not. The strongest journals show graph, algebra, context, units, limitations, and revision together.

MODELING HABITS

What strong modeling looks like.

This page is a guide for the kind of thinking the journal should make visible. It is not a checklist; it is a way to recognize stronger mathematical investigation.

Strong modeling

explains assumptions
critiques realism
revises predictions
compares models
identifies limitations
connects graph ↔ algebra ↔ context
interprets uncertainty

Weak modeling

only calculates
ignores unrealistic outputs
hides uncertainty
treats models as perfect truth
separates math from context
copies technology output without interpretation

Core idea

Models are arguments. They become stronger when evidence, assumptions, revisions, and limits become more visible.

USE MODEL

Shared field notebook. Individual accountability.

The project is collaborative, but grading is individual. The journal and individual evidence serve different purposes.

Physical group journal

The physical journal captures the group's evolving mathematical thinking. It should look used: sketched on, annotated, revised, taped into, crossed out, and questioned.

Individual digital evidence

Your individual digital evidence captures your own mathematical reasoning inside the group investigation. This may include personal graph notes, recalculations, interpretations, AI critiques, limitation reflections, and explanations of how your thinking changed.

How they work together

The group journal preserves the messy investigation trail. Individual evidence shows how each student reasoned, interpreted, revised, and contributed mathematically inside that shared work.

Field notebook rule

Clean pages are not the goal. Crossed-out thinking belongs here. Strong investigations often become more nuanced, not more certain.

RUBRIC ALIGNMENT

Use the journal to make standards visible.

The rubric does not only score final answers. It looks for reasoning, interpretation, revision, limitations, precision, and flexible thinking across the investigation.

1. Sequences and Series

Content Standard

2. Exponentials and Logarithms

Content Standard

3. Polynomials

Content Standard

4. Mathematical Communication and Interpretation

Skill Grade

5. Comparing Models, Assumptions, and Limitations

Skill Grade

6. Algebraic Precision and Flexible Number Sense

Skill Grade

CLASS 1

Entering the System

What relationships are we beginning to notice?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Modeling anchor

Models are arguments.

SYSTEM MAP CANVAS

Sketch variables, links, pressures, unknowns, and possible feedback loops.

student thinking space

Possible artifacts

system map, arrows between variables, uncertainty notes, pressure labels, feedback loop sketch

Reflection prompt

What assumption became weaker?

VARIABLE TABLE AND MEASUREMENT NOTES

Variable | Units | Why it matters | What is hard to measure. Which variables seem politically, socially, or physically difficult to measure?

student thinking space

Possible artifacts

variable table, units list, source notes, measurement concern, highlighted hard-to-measure variable

Reflection prompt

What evidence became more convincing?

Example thinking

Example thinking: Transportation time may be measurable, but informal delays or refrigeration failures may be harder to capture.

SKETCH BEFORE TECHNOLOGY

Predict the shape of one relationship before using Desmos, Sheets, or another tool. What relationship seems obvious at first but may be misleading?

student thinking space

Possible artifacts

rough graph sketch, predicted trend, estimated intercept, domain note, comparison to later Desmos graph

Reflection prompt

What hidden variable appeared?

Discuss uncertainty, early assumptions, possible misleading patterns, and variables that may be hard to measure reliably.

student thinking space

Possible artifacts

conversation bullets, changed assumption, unclear question, photo of journal page, next-step note

Reflection prompt

What still feels uncertain?

Make your own reasoning visible.

This is a small individual accountability moment inside the shared investigation. Choose one mathematical contribution you personally understand and can defend.

Possible focus

Explain one graph feature, critique one assumption, interpret one threshold, identify one limitation, explain one revision, or defend one modeling decision.

Evidence to attach or reference

graph annotation, recalculation, model critique, revised assumption note, scatterplot interpretation, limitation analysis, or synthesis explanation

Keep it personal

Write in your own mathematical voice. Do not divide the project by topic; every student should engage across the major lenses.

individual reasoning snapshot

CLASS 2

Modeling Change

How is the system changing over time?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Modeling anchor

Models reveal and hide.

DISCRETE VS CONTINUOUS REASONING

Does your context change in steps, continuously, or both? Justify the representation that best fits your system.

student thinking space

Possible artifacts

short paragraph, timeline, sampled data table, continuous curve sketch, units annotation

Reflection prompt

What graph feature mattered most?

PATTERN BEHAVIOR

Does the system behave additively, multiplicatively, unpredictably, or in phases? What evidence supports this?

student thinking space

Possible artifacts

difference/ratio table, highlighted trend, model choice note, evidence for add/multiply/phase behavior

Reflection prompt

What made the model less realistic?

Write useful formulas if the pattern supports them. Annotate what each term, parameter, and assumption means in your context.

student thinking space

Possible artifacts

recursive formula, explicit formula, variable definitions, annotated parameters, substitution example

Reflection prompt

What tradeoff emerged?

What does accumulated change represent in your system? Would accumulation realistically continue forever? Why or why not?

student thinking space

Possible artifacts

partial sum table, cumulative graph, written interpretation, finite/infinite note, realism critique

Reflection prompt

What would strengthen this claim?

Example thinking

Example thinking: Total loss can accumulate across storage and transport stages, but it cannot continue forever because the remaining food eventually reaches zero.

MEANINGFUL TIME SCALES

Choose multiple time ranges that matter for your investigation. How does the model behave differently across those scales?

student thinking space

Possible artifacts

short vs long range graph, table of predictions, reliability note, revised domain

Reflection prompt

What revision improved the model most?

When does the pattern stop behaving consistently? What evidence would make you revise or reject it?

student thinking space

Possible artifacts

crossed-out prediction, revised assumption, graph annotation, note explaining why pattern changed

Reflection prompt

What assumption became weaker?

CLASS 2 INDIVIDUAL SNAPSHOT

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STOP AND CRITIQUE

Pause before you keep building.

A strong investigation interrupts itself. Use this page to evaluate whether your current model is becoming more credible or just more polished.

Credibility questions

What currently feels misleading?
What assumption is dominating the model?
What variable is missing?
What evidence would weaken your claim?

Model honesty

Which prediction currently feels least realistic?
What would an opposing interpretation say?
Which graph feature might be overinterpreted?

Anchor

Models are arguments. Revision strengthens modeling when it makes assumptions, evidence, and limits clearer.

critique notes and revised decisions

END OF CLASS 2

System Plan + Model Connection Check-In

Ensure your system, variables, early sequence work, data, assumptions, and planned models connect into a coherent mathematical investigation before deeper modeling begins.

Students should show

chosen system; key variables; changing relationships; planned models; why each model makes sense; what each model reveals; what each may fail to capture; data sources; assumptions

Teacher feedback focus

meaningful variables; appropriate models; measurable context; realistic assumptions; whether the work is too broad, too shallow, or forcing functions unnaturally

Formative purpose

This is a low-stakes check-in for redirection, clarity, rigor, and coherence. The goal is support before final submission, not punishment.

checkpoint notes, feedback, and next revisions

Changing assumptions creates different mathematical realities.

Choose model versions that are meaningful for your context. Which model is more useful, more realistic, or more honest about uncertainty? Which model hides constraints, exaggerates certainty, or breaks first?

version A

version B

version C

CLASS 3

Acceleration, Decay, and Thresholds

When does repeated change become accelerated or compounding?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Modeling anchor

Assumptions shape predictions.

What evidence suggests the change depends on the current amount rather than a constant added amount?

student thinking space

Possible artifacts

linear/exponential graph comparison, residual or fit note, written explanation, trendline screenshot

Reflection prompt

What evidence became more convincing?

How does continuous exponential modeling change the interpretation compared to sampled or discrete terms in your context?

student thinking space

Possible artifacts

sequence table, continuous curve, interpretation note, domain or time-unit comparison

Reflection prompt

What hidden variable appeared?

BUILD THE MODEL

Define starting value, growth or decay factor, variables, assumptions, and realistic domain. Which assumption is doing the most work?

student thinking space

Possible artifacts

exponential equation, parameter labels, Desmos screenshot, assumption note, realistic domain

Reflection prompt

What still feels uncertain?

Use logarithms when you need to solve for an unknown time or threshold. What does that threshold mean in the real system?

student thinking space

Possible artifacts

threshold equation, logarithmic solving steps, graph verification, contextual interpretation

Reflection prompt

What graph feature mattered most?

RATE OR ASSUMPTION COMPARISON

Compare multiple realistic rates or assumptions relevant to your investigation. Which assumptions cause the prediction to change dramatically?

student thinking space

Possible artifacts

multiple-rate graph, comparison table, threshold shift note, revised prediction

Reflection prompt

What made the model less realistic?

MODEL BREAKDOWN

Where does exponential behavior stop being realistic? What real-world forces might interrupt it? What hidden variables matter?

student thinking space

Possible artifacts

limitation reflection, hidden-variable list, unrealistic prediction, interrupted-growth explanation

Reflection prompt

What tradeoff emerged?

Example thinking

Example thinking: A price model may keep rising mathematically, but consumer behavior, policy changes, or supply shifts could interrupt the trend.

CLASS 3 INDIVIDUAL SNAPSHOT

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Possible focus

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Evidence to attach or reference

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Anchor

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critique notes and revised decisions

CLASS 3 MODEL COMPARISON STUDIO

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version C

CLASS 4

Constraints, Capacity, and Cost

How do physical and economic constraints reshape the system?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Why polynomial structure belongs here

Physical systems have dimensions, limits, packing constraints, refrigeration volume, warehouse capacity, shipping tradeoffs, and cost interactions. Polynomial models help reveal how those interacting constraints reshape what is possible.

Modeling anchor

Revision strengthens modeling.

SYSTEM CONSTRAINT FRAMING

Before building a polynomial, identify the real constraint: storage capacity, reservoir volume, transportation packing, refrigeration space, warehouse limits, shipping optimization, or another physical tradeoff. Why does capacity matter in your system?

student thinking space

Possible artifacts

constraint map, storage/transport sketch, capacity note, cost or space tradeoff

Reflection prompt

What would strengthen this claim?

Example thinking

Example thinking: If food exists but cannot be stored cold, refrigeration volume becomes a system constraint rather than just a design detail.

FLAT BLUEPRINT

Original dimensions, square cutout variable, folding lines, units in centimeters. Connect the drawing to a real storage, packing, refrigeration, or distribution constraint.

student thinking space

Possible artifacts

labeled blueprint, dimensions, units, cutout variable, folding lines

Reflection prompt

What revision improved the model most?

FOLDED STORAGE STRUCTURE

Height, interior dimensions, usable space, loading floor. Explain what the structure represents in your food system.

student thinking space

Possible artifacts

folded sketch, interior dimensions, height annotation, usable capacity note

Reflection prompt

What assumption became weaker?

VOLUME POLYNOMIAL

Factored form, standard form, and why multiplying interacting dimensions creates a cubic relationship.

student thinking space

Possible artifacts

factored form, standard form, expanded work, explanation of interacting dimensions

Reflection prompt

What evidence became more convincing?

GRAPH ANNOTATION

Intercepts, multiplicity, extrema, domain, unrealistic regions, physical meaning.

student thinking space

Possible artifacts

annotated polynomial graph, intercept labels, extrema, domain restriction, physical interpretation

Reflection prompt

What hidden variable appeared?

Example thinking

Example thinking: A negative volume may appear on the graph, but it has no physical meaning for a storage system.

Degree, even/odd, leading coefficient, as $x \to \infty$ and as $x \to -\infty$.

student thinking space

Possible artifacts

degree note, leading coefficient explanation, $x \to \infty$ and $x \to -\infty$ interpretation

Reflection prompt

What still feels uncertain?

How could division help analyze storage, packing, capacity, grouping, or remaining space? Interpret the quotient and remainder physically.

student thinking space

Possible artifacts

long division work, quotient/remainder interpretation, grouping or remaining-space explanation

Reflection prompt

What graph feature mattered most?

CONSTRAINT REVISION

Change a realistic physical or economic constraint from your context. What tradeoff appears, and which graph regions become impossible?

student thinking space

Possible artifacts

revised dimensions, new graph, comparison note, tradeoff explanation, impossible-region annotation

Reflection prompt

What made the model less realistic?

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Possible focus

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Anchor

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critique notes and revised decisions

BEGINNING OF CLASS 4

Polynomial + Physical Constraints Check-In

Ensure the polynomial work is connected to physical constraints, not treated as an isolated polynomial unit.

Students should show

blueprint or sketch; dimensions; polynomial model; graph; annotations; early interpretation; domain reasoning

Teacher feedback focus

why the relationship is polynomial; why it is not linear or exponential; cubic vs quadratic behavior; turning points; unrealistic regions; long division; factor interpretation; model comparison

Formative purpose

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CLASS 5

Revision, Synthesis, and Communication

What story does our mathematics tell?

Field notebook expectation

Use this section as a place for rough work, uncertainty, recalculation, and revision. Let the thinking stay visible.

Modeling anchor

Correlation is not causation.

SYNTHESIS TABLE

Lens | What did it reveal? | Where did it fail?

student thinking space

Possible artifacts

lens table, reveal/fail notes, agreement/conflict annotations

Reflection prompt

What tradeoff emerged?

WHERE MODELS AGREED OR CONFLICTED

Compare mathematical lenses and assumptions.

student thinking space

Possible artifacts

comparison paragraph, color-coded model notes, assumptions table, contradiction note

Reflection prompt

What would strengthen this claim?

FINAL MATHEMATICAL CLAIM

Use evidence, assumptions, limitations, tradeoffs, and implications. What remained uncertain even after modeling?

student thinking space

Possible artifacts

claim draft, evidence list, limitation statement, implication note

Reflection prompt

What revision improved the model most?

COMMUNICATION STORYBOARD

Choose a format. Plan how mathematical evidence, revisions, and uncertainty will be visible.

student thinking space

Possible artifacts

format plan, journal-page sequence, graph order, speaking notes, evidence map

Reflection prompt

What assumption became weaker?

IF WE REPEATED THIS INVESTIGATION

With better data, what would change? Which assumption, measurement, or model would you revisit first?

student thinking space

Possible artifacts

better-data plan, source wishlist, revised measurement idea, next-model note

Reflection prompt

What evidence became more convincing?

INDIVIDUAL ARTIFACT SNAPSHOT

Revision, hidden-variable analysis, limitation critique, or r/R^2 interpretation. A high R^2 can still mislead; correlation can support a claim without proving causation.

student thinking space

Possible artifacts

personal graph note, individual critique, threshold interpretation, limitation analysis, revised assumption

Reflection prompt

What hidden variable appeared?

Example thinking

Example thinking: I supported changing the growth rate because the original assumption made the long-term prediction unrealistic.

CLASS 5 INDIVIDUAL SNAPSHOT

Make your own reasoning visible.

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Possible focus

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